Sturmian sequences of $\sqrt{n}$: the proofs

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This is the proof-dense companion to `sturmian.tex`. Every claim made there is proved here. Two claims that `sturmian.tex` states for $n = 2$ and $n = 7$ alone are proved for every n in §9, and one statement there is false as written; §6 gives the counterexample and the repair.

1 Codings

Fix an irrational $\alpha \in (0, 1)$. Write $\{y\}$ for $y - \lfloor y \rfloor$ and, for $x \in [0, 1)$,

$$x_k \;=\; \{x + k\alpha\}, \qquad k \geq 0.$$

Definition 1. The *lower* and *upper codings* of x are the sequences

$$\mathcal{C}(\alpha, x)_k \;=\; \begin{cases} 1 & x_k \in [1-\alpha,\, 1) \\ 0 & \text{otherwise,} \end{cases} \qquad \mathcal{C}^+(\alpha, x)_k \;=\; \begin{cases} 1 & x_k \in (1-\alpha,\, 1) \text{ or } x_k = 0 \\ 0 & \text{otherwise.} \end{cases}$$

The two codings read the same circle with the same rotation and differ only in which endpoint of the arc of length α carries the letter 1.

Lemma 2. *For $x \in [0, 1)$,*

$$\mathcal{C}(\alpha, x)_k \;=\; \lfloor (k+1)\alpha + x \rfloor - \lfloor k\alpha + x \rfloor, \qquad \mathcal{C}^+(\alpha, x)_k \;=\; \lceil (k+1)\alpha + x \rceil - \lceil k\alpha + x \rceil.$$

Proof. Put $m = \lfloor k\alpha + x \rfloor$ and $f = x_k$, so $k\alpha + x = m + f$ and $(k+1)\alpha + x = m + f + \alpha$ with $f + \alpha \in [0, 2)$. Hence $\lfloor (k+1)\alpha + x \rfloor - m = \lfloor f + \alpha \rfloor$, which is 1 exactly when $f \geq 1 - \alpha$.

For the second, $\lceil m + f \rceil = m$ when $f = 0$ and $m + 1$ otherwise, so $\lceil m + f + \alpha \rceil - \lceil m + f \rceil = 1$ exactly when $f = 0$ or $f + \alpha > 1$. □

So $\mathcal{C}(\alpha, x)$ is the sequence of `sturmian.tex` §2 with $\rho = x$.

Lemma 3. *$\mathcal{C}(\alpha, x)$ and $\mathcal{C}^+(\alpha, x)$ are equal unless $x_{k_0} = 0$ for some $k_0 \geq 0$, which happens for at most one k_0. In that case they differ exactly at the positions $k_0 - 1$ and k_0, where they read `10` and `01`.*

Proof. The letter rules differ only at $x_k \in \{0,\, 1-\alpha\}$. If $x_k = 1-\alpha$ then $x_{k+1} = 0$, and if $x_{k_0} = 0$ with $k_0 \geq 1$ then $x_{k_0-1} = 1 - \alpha$. Two indices with $x_k = 0$ would give $(j - k)\alpha \in \mathbb{Z}$. □

Lemma 4. *For $x \in [0, 1)$ and $N \geq 1$ the number of 1s among the first N letters of $\mathcal{C}(\alpha, x)$ is $\lfloor N\alpha + x \rfloor$.*

Proof. Sum Lemma 2 over $k < N$ and use $\lfloor x \rfloor = 0$. □

Corollary 5. *The density of 1s in $\mathcal{C}(\alpha, x)$ is α. The same holds for $\mathcal{C}^+(\alpha, x)$.*

Proof. $\lfloor N\alpha + x \rfloor / N \to \alpha$. The upper coding differs in at most two positions. □

The density is called the *slope*. It is a property of the sequence alone, so two codings that are equal as sequences have equal slopes. Every use of "solve for α" below rests on this corollary.

Lemma 6. *$\mathcal{C}(\alpha, x)_{k+1} = \mathcal{C}(\alpha, \{x+\alpha\})_k$, and likewise for $\mathcal{C}^+$.*

Proof. $\{x + (k+1)\alpha\} = \{\{x+\alpha\} + k\alpha\}$. □

2 The orbit is dense

Lemma 7. *For irrational α and any $x \in [0,1)$, the set $\{x_k : k \geq 0\}$ is dense in $[0,1)$.*

Proof. Let $\varepsilon > 0$. By Dirichlet there are integers $q \geq 1$ and p with $|q\alpha - p| < 1/q < \varepsilon$; put $\delta = |q\alpha - p| > 0$. The points x_0, x_q, x_{2q}, … differ consecutively by $\pm\delta$ modulo 1, with a fixed sign, so the first $\lceil 1/\delta \rceil + 1$ of them meet every interval of length δ. Take $\delta < \varepsilon$. □

Corollary 8. *$x \mapsto \mathcal{C}(\alpha, x)$ is injective on $[0,1)$.*

Proof. Let $x < y$ and suppose the codings agree. By Lemma 4, $\lfloor N\alpha + x \rfloor = \lfloor N\alpha + y \rfloor$ for every $N \geq 1$. Choose N with $\{N\alpha + x\} > 1 - (y - x)$; then $N\alpha + x < \lceil N\alpha + x \rceil \leq N\alpha + y$. □

3 Complexity

Definition 9. A *factor* of length m of a sequence ℓ is a block $\ell_k \ell_{k+1} \cdots \ell_{k+m-1}$. Let $F_\ell(m)$ be the number of distinct factors of length m. The sequence is *Sturmian* when $F_\ell(m) = m + 1$ for every $m \geq 1$.

Theorem 10. *$F_{\mathcal{C}(\alpha,x)}(m) = m + 1$ for every $m \geq 1$.*

Proof. For $j \geq 1$ and $t \in [0,1)$,

$$\lfloor t + j\alpha \rfloor \;=\; \lfloor j\alpha \rfloor + \begin{cases} 1 & t \geq \{-j\alpha\} \\ 0 & t < \{-j\alpha\}, \end{cases}$$

because $t + j\alpha = \lfloor j\alpha \rfloor + (\{j\alpha\} + t)$ with $\{j\alpha\} + t \in [0,2)$, and $\{j\alpha\} + t \geq 1$ exactly when $t \geq 1 - \{j\alpha\} = \{-j\alpha\}$.

Let w be the factor of length m at position k. By Lemma 6 its letters are $w_j = \mathcal{C}(\alpha, x_k)_j$ for $j < m$, so by Lemma 4 the partial sums of w give $\lfloor x_k + j\alpha \rfloor$ for every $j \leq m$. By the display, w therefore determines, for each $j \in \{1, \ldots, m\}$, whether $x_k \geq \{-j\alpha\}$; and conversely those m comparisons determine w.

The m points $\{-j\alpha\}$, $1 \leq j \leq m$, are distinct and lie in $(0,1)$, so they cut $[0,1)$ into $m+1$ intervals of positive length. Factors of length m are in bijection with those intervals that contain some x_k, and by Lemma 7 every one of them does. □

Corollary 11. *$F_{\mathcal{C}^+(\alpha,x)}(m) = m + 1$ for every $m \geq 1$.*

Proof. One can rerun the proof above with $\lceil \cdot \rceil$ for $\lfloor \cdot \rfloor$: the comparisons become $x_k > \{-j\alpha\}$ and the intervals are half-open at the other end. But that is more work than is needed, because an upper coding is the bitwise complement of a lower coding:

$$\mathcal{C}^+(\alpha, x)_k + \mathcal{C}(1-\alpha, -x)_k = 1 \qquad (k \geq 0).$$

Indeed $\mathcal{C}^+(\alpha,x)_k = 1$ exactly when $\{x+k\alpha\} \in (1-\alpha,1)\cup\{0\}$, whose complement in $[0,1)$ is $(0,1-\alpha]$; and $\{-x+k(1-\alpha)\} = \{-(x+k\alpha)\}$, so the reflection $v \mapsto 1-v$ sends the orbit of x at slope α to the orbit of $-x$ at slope $1-\alpha$, carrying $(0,1-\alpha]$ onto $[\alpha,1) = [1-(1-\alpha),1)$ — the arc of the lower coding of slope $1-\alpha$. The exceptional point $v=0$ is fixed by the reflection and lies outside $[\alpha,1)$, as it must. Complementation is a bijection on words, and $1-\alpha$ is again irrational and in $(0,1)$, so Theorem 10 gives the count. □

The second proof is the one formalised. In `lean/Sturmian.lean` the displayed identity is `codeUp_add_code_reflect`, and the corollary is `ncard_factors_codeUp`. The first proof would have cost a parallel development: the natural range of the base point for an upper coding is $(0,1]$ rather than $[0,1)$, since $\mathcal{C}^+$ reads $\{x+k\alpha\}=0$ as the letter 1.

For the Fibonacci sequence of `sturmian.tex` §1 and $m=3$ the four intervals are cut by $\{-\alpha\},\{-2\alpha\},\{-3\alpha\}$ with $\alpha = 1/\varphi$, and the four factors are `101`, `011`, `110`, `010`. That is the table in §1.

4 Gaps

Write

$$\frac{1}{\alpha} \;=\; a_1+\beta, \qquad a_1 = \lfloor 1/\alpha \rfloor \geq 1, \qquad \beta = \{1/\alpha\} \in (0,1).$$

Then $a_1\alpha < 1 < (a_1+1)\alpha$, which is the inequality of `sturmian.tex` §3.

Let $I = [1-\alpha,1)$, the arc carrying the letter 1, and coordinatise it by

$$x \;=\; (1-\alpha)+\alpha u, \qquad u \in [0,1).$$

Theorem 12. *Let $x \in I$ with coordinate u. The least $t \geq 1$ with $\{x+t\alpha\} \in I$ is*

$$t \;=\; \begin{cases} a_1+1 & u < \beta \\ a_1 & u \geq \beta, \end{cases}$$

and the coordinate of $\{x+t\alpha\}$ is $\{u+1-\beta\}$.

Proof. $x+\alpha \in [1,1+\alpha)$, so $\{x+\alpha\} = x+\alpha-1 = \alpha u$, and for $s \geq 1$,

$$\{x+s\alpha\} \;=\; \alpha(u+s-1) \qquad \text{as long as } \alpha(u+s-1) < 1.$$

Now $\alpha(u+s-1) < 1 \iff s-1 < 1/\alpha - u = a_1+\beta-u$, and $\alpha(u+s-1) \geq 1-\alpha \iff u+s \geq a_1+\beta$. So $\{x+s\alpha\} \in I$ exactly when

$$a_1+\beta-u \;\leq\; s \;<\; a_1+\beta-u+1,$$

whose unique integer solution is $s = \lceil a_1+\beta-u \rceil$. For $u < \beta$ that is a_1+1 and for $u \geq \beta$ it is a_1; in both cases s is the least such t because the displayed formula for $\{x+s\alpha\}$ is valid and below $1-\alpha$ for all smaller s.

The new coordinate is $(\alpha(u+t-1)-(1-\alpha))/\alpha = u+t-1/\alpha$, which is $u+1-\beta$ when $t = a_1+1$ and $u-\beta$ when $t = a_1$. □

Corollary 13. *In $\mathcal{C}(\alpha,x)$ the gap between consecutive 1s is a_1 or a_1+1, so the number of 0s between them is a_1-1 or a_1.*

Definition 14. Let $k_0 < k_1 < k_2 < \cdots$ be the positions of the 1s of ℓ. The *derived sequence* ℓ' is

$$\ell'_i \;=\; \begin{cases} 1 & a_1 \text{ zeros lie between the } i\text{th and } (i{+}1)\text{th 1} \\ 0 & a_1 - 1 \text{ do.} \end{cases}$$

By Theorem 12, $\ell'_i = 1$ exactly when the coordinate u_i of x_{k_i} satisfies $u_i < \beta$, and

$$u_{i+1} \;=\; \{u_i + 1 - \beta\}.$$

Lemma 15 (block decomposition)**.** *Define* $\tau_a(1) = \mathtt{1}\,\mathtt{0}^a$ *and* $\tau_a(0) = \mathtt{1}\,\mathtt{0}^{a-1}$*. Then*

$$\ell \;=\; \mathtt{0}^{k_0} \cdot \tau_{a_1}(\ell').$$

Proof. The ith block of ℓ starting at k_i is a 1 followed by the zeros before the next 1, of which there are a_1 if $\ell'_i = 1$ and $a_1 - 1$ if $\ell'_i = 0$. □

Deriving and τ_{a_1} are inverse to each other up to the prefix $\mathtt{0}^{k_0}$. That prefix is the only reason a shift appears anywhere below.

5 The derivation theorem

Theorem 16. *Let* u_0 *be the coordinate of* x_{k_0}*, the first* 1*. Then*

$$\mathcal{C}(\alpha, x)' \;=\; \mathcal{C}^+(\beta,\, \{-u_0\}), \qquad\qquad \mathcal{C}^+(\alpha, x)' \;=\; \mathcal{C}(\beta,\, 1 - u_0).$$

Proof. For the lower coding put $v_i = \{-u_i\}$. By Theorem 12, $v_{i+1} = \{v_i + \beta\}$, so $v_i = \{v_0 + i\beta\}$. And

$$\ell'_i = 1 \iff u_i \in [0, \beta) \iff -u_i \in (-\beta, 0] \iff v_i \in (1 - \beta, 1) \text{ or } v_i = 0,$$

which is the letter rule of $\mathcal{C}^+(\beta, v_0)$.

For the upper coding the arc is $I^+ = (1 - \alpha, 1]$, coordinatised by $u \in (0, 1]$ with $u = 1$ meaning $x \equiv 0$. The computation of Theorem 12 is unchanged except that $\{x + s\alpha\} \in I^+$ reads $u + s > a_1 + \beta$ and $u + s - 1 \le a_1 + \beta$, so $t = a_1 + 1$ exactly when $u \le \beta$. Put $v_i = 1 - u_i \in [0, 1)$; then $v_{i+1} = \{v_i + \beta\}$ and

$$\ell'_i = 1 \iff u_i \le \beta \iff v_i \ge 1 - \beta,$$

which is the letter rule of $\mathcal{C}(\beta, v_0)$. □

Deriving sends the slope α to $\beta = \{1/\alpha\}$ and exchanges the two codings. The exchange is invisible for all but countably many x (Lemma 3), and it is exactly what makes the statement of Theorem 16 an equality rather than an equality off a small set.

Corollary 17. *The derived sequence of a coding is a coding, hence Sturmian, and its slope is* $\{1/\alpha\}$*.*

Corollary 18. *The slopes of* $\ell, \ell', \ell'', \ldots$ *are the orbit of* α *under the Gauss map* $\alpha \mapsto \{1/\alpha\}$*, so the block widths* a_1 *encountered are the partial quotients of* $\alpha = [\,0;\, a_1, a_2, a_3, \ldots\,]$*.*

6 Morse and Hedlund

One direction is Theorem 10: every coding with irrational slope is Sturmian. The statement in `sturmian.tex` §2 asserts the converse in the form

$$\ell_k \;=\; \lfloor (k+1)\alpha+\rho\rfloor - \lfloor k\alpha+\rho\rfloor, \qquad \rho\in[0,1),$$

that is, $\ell = \mathcal{C}(\alpha,\rho)$. In that form it is false.

Let $\alpha = 1/\varphi = (\sqrt{5}-1)/2$ and let $c = \mathcal{C}(\alpha,\alpha)$, the Fibonacci sequence. Let $\mathtt{1}c$ be c with a 1 prefixed.

Theorem 19. *$\mathtt{1}c = \mathcal{C}^+(\alpha,0)$. It is Sturmian, and there is no $x\in[0,1)$ with $\mathtt{1}c = \mathcal{C}(\alpha',x)$ for any irrational $\alpha'\in(0,1)$.*

Proof. $\mathcal{C}^+(\alpha,0)_0 = 1$ because $x_0 = 0$, and $\mathcal{C}^+(\alpha,0)_{k+1} = \mathcal{C}^+(\alpha,\alpha)_k = \mathcal{C}(\alpha,\alpha)_k$ by Lemma 6 and Lemma 3, the orbit of α missing 0. So $\mathtt{1}c = \mathcal{C}^+(\alpha,0)$, and it is Sturmian by Corollary 11.

By Lemma 4 the first $N-1$ letters of c carry $\lfloor (N-1)\alpha+\alpha\rfloor = \lfloor N\alpha\rfloor$ ones, so the first N letters of $\mathtt{1}c$ carry $1+\lfloor N\alpha\rfloor$.

Suppose $\mathtt{1}c = \mathcal{C}(\alpha',x)$. The slopes agree by Corollary 5, so $\alpha' = \alpha$, and Lemma 4 gives $\lfloor N\alpha + x\rfloor = 1+\lfloor N\alpha\rfloor$ for every $N\ge 1$. Since $x\in[0,1)$ this forces $\{N\alpha\}+x\ge 1$ for every $N\ge 1$. By Lemma 7 there is N with $\{N\alpha\} < 1-x$. □

The repair is to admit both codings.

Theorem 20 (Morse and Hedlund)**.** *A sequence is Sturmian if and only if it is $\mathcal{C}(\alpha,x)$ or $\mathcal{C}^+(\alpha,x)$ for some irrational $\alpha\in(0,1)$ and some $x\in[0,1)$.*

The forward implication is Theorem 10 and Corollary 11. For the converse, run §4 and §5 backwards. Steps 1 to 4 are below; step 5 is §7.

1. $F_\ell(1) = 2$, so both letters occur. $F_\ell(2) = 3$, so exactly one word of length 2 is absent, and it is $\mathtt{00}$ or $\mathtt{11}$: were $\mathtt{01}$ absent, every 1 would precede every 0 and $\ell = \mathtt{1}^p\mathtt{0}^\infty$, whose complexity is $p+1$ for all $m > p$. *Proved.*
2. The lengths of the maximal blocks of 0s take exactly two values, and they are consecutive, say $a-1$ and a. *Proposition 23 below.*
3. Given step 2, Lemma 15 applies verbatim: ℓ factors as $\mathtt{0}^{k_0}\tau_a(\ell')$ with ℓ' a sequence over $\{\mathtt{0},\mathtt{1}\}$. *Proved.*
4. ℓ' is again Sturmian. *Proposition 28 below*; that both of its letters occur is the case $m = 1$.
5. Iterating pins a single α and a single x. *§7:* Lemma 29 and Theorem 33.

Step 1 delivers $\mathtt{00}$ or $\mathtt{11}$, and step 2 is stated for $\mathtt{11}$. Exchanging the letters costs nothing: by the identity in the proof of Corollary 11, the complement of $\mathcal{C}(\beta,y)$ is $\mathcal{C}^+(1-\beta,-y)$, and Theorem 20 admits both codings.

Step 2 is a counting argument. Call a factor u of ℓ *right-special* when $u\mathtt{0}$ and $u\mathtt{1}$ are both factors.

Lemma 21. *$F_\ell(m+1) - F_\ell(m)$ is the number of right-special factors of length m.*

Proof. Every factor of length $m+1$ is ua for a factor u of length m and a letter a, and every factor of length m has at least one right extension: it occurs at some k, and ℓ_{k+m} extends it. So $ua\mapsto u$ is onto the factors of length m, with two elements over u exactly when u is right-special. □

Corollary 22. *$F_\ell(m) = m + 1$ for every m if and only if ℓ has exactly one right-special factor of each length.*

The pair that does the work is $\mathtt{0}^{j+1}$ and $\mathtt{10}^{j}$. They have the same length and differ in their first letter, so at most one of them is right-special.

Proposition 23. *Let ℓ be Sturmian with $\mathtt{11}$ not a factor, and let a be the length of the longest block of $\mathtt{0}$s. Then $a \geq 2$, the blocks of $\mathtt{0}$s between consecutive $\mathtt{1}$s have lengths $a - 1$ and a and both occur, and the block before the first $\mathtt{1}$ has length at most a.*

Proof. ℓ has infinitely many $\mathtt{1}$s: otherwise it ends in $\mathtt{0}^\infty$ and F_ℓ is bounded. Let

$$S \;=\; \{\, j \;:\; \mathtt{10}^{j}\mathtt{1} \text{ is a factor of } \ell \,\},$$

the set of interior block lengths. S is nonempty, and $j \geq 1$ for $j \in S$ because $\mathtt{11}$ is absent.

The blocks of $\mathtt{0}$s are bounded. Were they not, S would be unbounded, only one block being initial. Take $j \in S$ and then $s \in S$ with $s \geq j + 1$. Now $\mathtt{10}^{j}\mathtt{1}$ and $\mathtt{10}^{j+1}$ are factors, so $\mathtt{10}^{j}$ is right-special; and $\mathtt{0}^{j+2}$ and $\mathtt{0}^{j+1}\mathtt{1}$ are factors, so $\mathtt{0}^{j+1}$ is right-special. Two of them, of length $j + 1$, against Corollary 22.

So a is defined. Every block is finite, so a block of length $\geq j$ is followed by a $\mathtt{1}$: hence $\mathtt{0}^{j}$ is a factor exactly for $j \leq a$, and $\mathtt{0}^{j}\mathtt{1}$ is a factor for $1 \leq j \leq a$.

Let $j \in S$ with $j < \max S$. Then $\mathtt{10}^{j}\mathtt{1}$ and $\mathtt{10}^{j+1}$ are factors, so $\mathtt{10}^{j}$ is right-special. If $j \leq a - 2$ then $\mathtt{0}^{j+2}$ and $\mathtt{0}^{j+1}\mathtt{1}$ are factors, so $\mathtt{0}^{j+1}$ is right-special as well, and Corollary 22 is contradicted. So

$$j \in S,\; j < \max S \implies j \geq a - 1.$$

If $\max S < a$, every $j \in S$ below $\max S$ would satisfy $j \geq a - 1 \geq \max S$, so there is none and $S = \{\max S\}$. A single block length makes ℓ eventually periodic, which bounds F_ℓ. So $\max S = a$, and the display gives $S \subseteq \{a-1, a\}$. Again S is not a single value, so $S = \{a-1, a\}$, and $a - 1 \geq 1$.

The initial block has length at most a by the definition of a. □

Proposition 23 is weaker than balancedness and is what steps 3 to 5 use. The classical route proves balancedness in full, by a minimal-counterexample induction producing a palindrome w with $\mathtt{0}w\mathtt{0}$ and $\mathtt{1}w\mathtt{1}$ both factors; that is not needed here.

Step 4 needs three more facts, all about counting, and then one bridge.

Lemma 24. *An eventually periodic sequence has bounded factor complexity.*

Proof. With preperiod q and period p, every factor is a factor at some position below $q + p$, so $F_\ell(n) \leq q + p$. □

Lemma 25. *If ℓ has no right-special factor of length m, it is eventually periodic.*

Proof. Every factor of length m has exactly one right extension, so the window $u_k = \ell_k \cdots \ell_{k+m-1}$ determines u_{k+1}. The windows lie in a finite set, so $u_0, u_1, \ldots$ is eventually periodic, and ℓ is read off its windows. □

Lemma 26. *If ℓ has exactly one right-special factor of each length, each is a suffix of the next.*

Proof. The suffix of length m of a right-special factor of length $m + 1$ is right-special. □

Lemma 27. *Let ℓ, a and ℓ' be as in step 3. Then w is a right-special factor of ℓ' if and only if $\tau_a(w)\mathtt{10}^{a-1}$ is a right-special factor of ℓ.*

Proof. The 1s of ℓ are exactly the block starts, and $\tau_a(w)$ begins with $\mathtt{1}$, so every occurrence of $\tau_a(w)$ in ℓ begins at a block start. So $w\mathtt{0}$ is a factor of ℓ' exactly when $\tau_a(w)\mathtt{10}^{a-1}\mathtt{1}$ is a factor of ℓ, and $w\mathtt{1}$ exactly when $\tau_a(w)\mathtt{10}^a$ is one, no block being longer than a. □

Proposition 28. *Let ℓ, a and ℓ' be as in step 3. Then ℓ' is Sturmian.*

Proof. At most one right-special factor of each length. Let w and w' be right-special factors of ℓ' of length m. By Lemma 27 both $\tau_a(w)\mathtt{10}^{a-1}$ and $\tau_a(w')\mathtt{10}^{a-1}$ are right-special factors of ℓ, so by Lemma 26 one is a suffix of the other, and after dropping the common $\mathtt{10}^{a-1}$ so is one of $\tau_a(w)$, $\tau_a(w')$. A suffix of $\tau_a(w')$ beginning with $\mathtt{1}$ begins at a block start, so it is $\tau_a(v)$ for a suffix v of w'; counting blocks, $v = w'$. So $\tau_a(w) = \tau_a(w')$ and $w = w'$.

At least one right-special factor of each length. Were there none of length m, ℓ' would be eventually periodic by Lemma 25, hence so would $\ell = \mathtt{0}^{k_0}\tau_a(\ell')$, and F_ℓ would be bounded by Lemma 24.

Corollary 22 now gives $F_{\ell'}(m) = m + 1$. □

7 Step 5

Normalise so that $\mathtt{11}$ is absent: by step 1 and the identity in the proof of Corollary 11, the complement of $\mathcal{C}(\beta, y)$ is $\mathcal{C}^+(1-\beta, -y)$, and both codings are admitted. Then Proposition 23 gives $a \geq 2$, and every level below is normalised the same way.

Write $\Delta_\alpha(u) = |u|_1 - \alpha|u|$ for a finite word u, where $|u|_1$ counts the 1s.

Lemma 29. *Let $a_1, a_2, \ldots \geq 2$ be the block parameters of the successive derived sequences and $\varepsilon_1, \varepsilon_2, \ldots \in \{0, 1\}$ record which levels were complemented. There is exactly one $\alpha \in (0, 1)$ compatible with this data, and it is irrational.*

Proof. A slope and the slope below it are related by $\alpha = 1/(a_1 + \beta)$ and $\beta \mapsto \beta$ or $1 - \beta$. On $[0, 1]$ the map $\beta \mapsto 1/(a_1 + \beta)$ has derivative $-1/(a_1 + \beta)^2$ of modulus at most $1/a_1^2 \leq 1/4$, and $\beta \mapsto 1 - \beta$ is an isometry. So the first j levels confine α to an interval of length at most 4^{-j}.

Were $\alpha = p/q$ in lowest terms, then $\beta = q/p - a_1$ has denominator $p < q$, and complementing does not change a denominator. Denominators cannot descend forever, and every level has a block parameter by Proposition 28. □

Lemma 30. $\Delta_\alpha(\tau_a(w)) = -\alpha\,\Delta_\beta(w)$.

Proof. $|\tau_a(w)| = a|w| + |w|_1$ and $|\tau_a(w)|_1 = |w|$, so

$$\Delta_\alpha(\tau_a(w)) = |w|(1 - a\alpha) - \alpha|w|_1 = \alpha\beta|w| - \alpha|w|_1.$$

□

Proposition 31. $|\Delta_\alpha(u)| < 1$ *for every factor u of ℓ.*

Proof. By induction on $|u|$, over all levels at once; α and β are the slopes of ℓ and ℓ', and $1/\alpha = a + \beta$.

Write $u = \mathtt{0}^s\,\tau_a(u')\,q$ with u' a factor of ℓ' and q empty or $q = \mathtt{1}\mathtt{0}^t$ a *proper* prefix of its block. A block has $a-1$ or a zeros, so $s \le a$ and $t \le a-1$.

For u' empty, $\Delta_\alpha(u) = -\alpha s$ or $1-\alpha(s+1+t)$, and $\alpha a < 1$ bounds both: $-1 < -\alpha a \le -\alpha s \le 0$ and $-1 < 1 - 2\alpha a \le 1-\alpha(s+1+t) \le 1-\alpha$.

For u' nonempty, $s = a$ forces the block to the left of u' to be the long one, so $\mathtt{1}u'$ is a factor of ℓ'. Also

$$|u| \;\ge\; s + a|u'| + |u'|_1 \;\ge\; a(1+|u'|) \;>\; |u'|+1,$$

so the induction hypothesis applies to u' and to $\mathtt{1}u'$. By Lemma 30,

$$\Delta_\alpha(u) \;=\; -\alpha s - \alpha\Delta_\beta(u') \;+\; \begin{cases} 1-\alpha(1+t) & q \neq \epsilon \\ 0 & q = \epsilon. \end{cases}$$

For the upper bound, $\Delta_\beta(u') > -1$ gives $-\alpha s - \alpha\Delta_\beta(u') < \alpha$ when $q = \epsilon$, and $s+1+t+\Delta_\beta(u') > 0$ gives $\Delta_\alpha(u) < 1$ when $q \neq \epsilon$.

For the lower bound, $s = a$ gives $\Delta_\beta(\mathtt{1}u') = 1 - \beta + \Delta_\beta(u') < 1$, hence $\Delta_\beta(u') < \beta$. Now:

$$\begin{array}{llll} q=\epsilon, & s \le a-1: & \alpha(s+\Delta_\beta(u')) < \alpha a & < 1 \\ q=\epsilon, & s = a: & \alpha(a+\beta) & = 1 \\ q\neq\epsilon, & s \le a-1: & \alpha(s+1+t+\Delta_\beta(u')) < 2\alpha a & < 2 \\ q\neq\epsilon, & s = a: & \alpha(2a+\beta) = \alpha a + 1 & < 2. \end{array}$$

□

Lemma 32. *The number of* 1*s among the first* N *letters of* $\mathcal{C}^+(\alpha, x)$ *is* $\lceil N\alpha + x\rceil - \lceil x\rceil$.

Proof. Sum Lemma 2 over $k < N$. □

Let $S(N)$ be the number of 1s among the first N letters of ℓ, and

$$D(N) \;=\; S(N) - N\alpha, \qquad D(0) = 0.$$

For $M < N$, $D(N) - D(M) = \Delta_\alpha(u)$ for the factor $u = \ell_M \cdots \ell_{N-1}$, so Proposition 31 says the values of D lie in an interval of length at most 1.

Theorem 33. *Put* $x = \sup_{N\ge 0} D(N)$. *Then* $x \in [0,1]$, *and*

$$\ell \;=\; \begin{cases} \mathcal{C}(\alpha, x) & \textit{if the supremum is attained} \\ \mathcal{C}^+(\alpha, x) & \textit{if not, and } x < 1 \\ \mathcal{C}^+(\alpha, 0) & \textit{if not, and } x = 1. \end{cases}$$

Proof. $D(0) = 0$ gives $x \ge 0$, and $D(N) - D(0) < 1$ gives $x \le 1$.

Attained at N_0: then $D(N) \le x$ and $x - D(N) = D(N_0) - D(N) < 1$ for every N, so $S(N) \le N\alpha + x < S(N) + 1$ and $\lfloor N\alpha + x\rfloor = S(N)$. Lemma 2 gives $\mathcal{C}(\alpha, x)_k = S(k+1) - S(k) = \ell_k$.

Not attained: then $D(N) < x$ for every N, and $x \le D(N) + 1$. For $x \in (0,1)$ this reads $S(N) < N\alpha + x \le S(N) + 1$, so $\lceil N\alpha + x\rceil = S(N) + 1$ and Lemma 32 applies with $\lceil x\rceil = 1$. For $x = 1$ it reads $S(N) \ge N\alpha$ and $S(N) < N\alpha + 1$, so $\lceil N\alpha\rceil = S(N)$. □

Together with Lemma 29 this is step 5, and Theorem 20 is proved. The sequence $\mathtt{1}c$ of Theorem 19 is the third case: its $D(N) = 1 - \{N\alpha\}$ has supremum 1, attained nowhere.

8 The substitution

For $a_1 = 1$ the two blocks are $\tau_1(1) = \texttt{10}$ and $\tau_1(0) = \texttt{1}$, so τ_1 is the substitution σ of `sturmian.tex` §1:

$$\sigma(1) = \texttt{10}, \qquad \sigma(0) = \texttt{1}.$$

Theorem 34. *Let $\alpha = 1/\varphi$ and $c = \mathcal{C}(\alpha, \alpha)$. Then $c' = c$, $\sigma(c) = c$, and $c = \lim_j \sigma^j(\texttt{1})$.*

Proof. $1/\alpha = \varphi = 1 + 1/\varphi$, so $a_1 = 1$ and $\beta = \alpha$. The first 1 of c is at $k_0 = 0$, where $x_0 = \alpha$ has coordinate $u_0 = (\alpha - (1 - \alpha))/\alpha = 2 - 1/\alpha = 1 - \alpha$. So $\{-u_0\} = \alpha$ and Theorem 16 gives $c' = \mathcal{C}^+(\alpha, \alpha) = \mathcal{C}(\alpha, \alpha) = c$, the middle equality by Lemma 3. Lemma 15 with $k_0 = 0$ then reads $c = \tau_1(c') = \sigma(c)$.

$\sigma(\texttt{1}) = \texttt{10}$ begins with `1`, and if $\sigma^j(\texttt{1})$ is a prefix of $\sigma^{j+1}(\texttt{1})$ then applying σ keeps it so; hence the limit exists. It is a prefix of every σ-fixed sequence beginning with `1`, and c is one. □

Theorem 34 is `fib_derive`, `sigma_pre` and `fib_limit` in `lean/Sturmian.lean`. Two remarks on the formalisation. First, $c' = c$ is the instance at the fixed point $\beta = \alpha$ of the general `step_low`, so the only new computation is $1/\alpha = 1 + \alpha$. Second, $\sigma(c) = c$ and $c = \lim_j \sigma^j(\texttt{1})$ are stated on prefixes — σ applied to the first m letters of c gives the first k_m letters of c, where k_m is the position of the mth 1 — because σ of an infinite sequence is not a function of position without first inverting the block-start map. Lemma 15 is `block_length` and `block_zero`, also positional.

The converse computation of `sturmian.tex` §6 is now legitimate. Suppose ℓ is a coding whose blocks are `1` and `10`, so $a_1 = 1$, and suppose $\ell' = \ell$. By Corollary 5 the slopes agree, so $\beta = \alpha$, and

$$\frac{1}{\alpha} \;=\; 1 + \alpha, \qquad \alpha^2 + \alpha - 1 = 0, \qquad \alpha = \frac{\sqrt{5} - 1}{2}.$$

9 Square roots

Write $\alpha_0 = \alpha$ and $\alpha_{j+1} = \{1/\alpha_j\}$ for the tails of the continued fraction, and $a_{j+1} = \lfloor 1/\alpha_j \rfloor$ for its partial quotients. Let $\ell^{(j)}$ be the jth derived sequence of ℓ.

Theorem 35. *Let $\ell = \mathcal{C}(\alpha, 0)$. For every $j \geq 1$,*

$$\ell^{(j)} \;=\; \begin{cases} \mathcal{C}^+(\alpha_j,\, \alpha_j) & j \text{ odd} \\ \mathcal{C}(\alpha_j,\, \alpha_j) & j \text{ even.} \end{cases}$$

Proof. Let γ be irrational in $(0, 1)$, $a = \lfloor 1/\gamma \rfloor$ and $\gamma' = \{1/\gamma\}$.

In $\mathcal{C}(\gamma, 0)$ the first 1 is at $k_0 = a$: for $k \leq a$ we have $k\gamma \leq a\gamma < 1$ so $x_k = k\gamma$, and $k\gamma \geq 1 - \gamma$ holds first at $k = a$ because $(a + 1)\gamma > 1$. Its coordinate is

$$u_0 \;=\; \frac{a\gamma - (1 - \gamma)}{\gamma} \;=\; a + 1 - \frac{1}{\gamma} \;=\; 1 - \gamma',$$

so $\{-u_0\} = \gamma'$ and Theorem 16 gives $\mathcal{C}(\gamma, 0)' = \mathcal{C}^+(\gamma', \gamma')$. With $\gamma = \alpha$ this is the case $j = 1$.

In $\mathcal{C}(\gamma, \gamma)$ and in $\mathcal{C}^+(\gamma, \gamma)$ the first 1 is at $k_0 = a - 1$, by the same count applied to $x_k = (k+1)\gamma$; the value $x_{k_0} = a\gamma$ and the coordinate $u_0 = 1 - \gamma'$ are as above. So Theorem 16 gives

$$\mathcal{C}(\gamma, \gamma)' = \mathcal{C}^+(\gamma', \gamma'), \qquad \mathcal{C}^+(\gamma, \gamma)' = \mathcal{C}(\gamma', \gamma'),$$

the second because $1 - u_0 = \gamma'$. Induction on j. □

Corollary 36. *If the continued fraction of α is purely periodic with period L, then $\ell = \mathcal{C}(\alpha, 0)$ satisfies*

$$\ell^{(L)} \;=\; \mathcal{C}(\alpha,\alpha) \;=\; \ell_1\ell_2\ell_3\cdots,$$

the sequence ℓ with its first letter removed.

Proof. $\alpha_L = \alpha$, so Theorem 35 gives $\mathcal{C}(\alpha,\alpha)$ or $\mathcal{C}^+(\alpha,\alpha)$, and those are equal by Lemma 3 because the orbit $\{(k+1)\alpha\}$ never meets 0. Lemma 6 identifies $\mathcal{C}(\alpha,\alpha)$ with the shift of ℓ. □

Corollary 37. *Conversely, if $\ell^{(L)}$ is a shift of ℓ then $\alpha_L = \alpha$, so the continued fraction of α is purely periodic with period dividing L.*

Proof. Shifting does not change the density, so Corollary 5 gives $\alpha_L = \alpha$. □

Now take n a non-square, $a_0 = \lfloor\sqrt{n}\rfloor$ and $\alpha = \sqrt{n} - a_0$. If $\sqrt{n} = [\,a_0;\ \overline{a_1,\ldots,a_L}\,]$ then $\alpha = [\,0;\ \overline{a_1,\ldots,a_L}\,]$, purely periodic, since the tails of α are the tails of $\sqrt{n}$ from index 1. Pure periodicity is Legendre's theorem, formalised in `lean/Palindrome.lean` as `purely_periodic`.

Corollary 38. *For every non-square n, the sequence $\ell = \mathcal{C}(\sqrt{n} - a_0,\, 0)$ returns to itself with one letter removed after exactly L derivations, where L is the period of the continued fraction of $\sqrt{n}$.*

`sturmian.tex` observes this for $n = 2$ and $n = 7$. For $n = 2$, $\alpha = \sqrt{2} - 1$, $a_1 = 2$, $\beta = \alpha$ and $L = 1$: one derivation returns the shift. For $n = 7$, $\alpha = \sqrt{7} - 2$ and the four tails are

$$\sqrt{7} - 2, \quad \frac{\sqrt{7}-1}{3}, \quad \frac{\sqrt{7}-1}{2}, \quad \frac{\sqrt{7}-2}{3}, \quad \sqrt{7} - 2,$$

with partial quotients $1, 1, 1, 4$, which is the table of §5 there. Every row of that table is Sturmian by Corollary 17 and Theorem 10. The intercept of each row equals its own slope, by Theorem 35.

10 Lagrange

Theorem 39. *If the continued fraction of α is eventually periodic then α is a quadratic irrational.*

Proof. Let $\alpha_m = \alpha_{m+L}$. Each tail is a Möbius image of the next, $\alpha_j = 1/(a_{j+1} + \alpha_{j+1})$, so composing L of them gives integers p, q, r, s with

$$\alpha_m \;=\; \frac{p\,\alpha_{m+L} + q}{r\,\alpha_{m+L} + s} \;=\; \frac{p\,\alpha_m + q}{r\,\alpha_m + s},$$

hence $r\alpha_m^2 + (s-p)\alpha_m - q = 0$ with $(r, s-p, q) \neq 0$. So α_m is a quadratic irrational, and composing m further Möbius maps with integer coefficients keeps α in $\mathbb{Q}(\alpha_m)$. □

This is `lean/Lagrange.lean`, where it is stated for an abstract sequence of tails, so it needs a witness: `Sturmian.sqrt_quadratic` in `lean/SqrtN.lean` supplies one. The Lean proves slightly more than the display above. The leading coefficient r is not merely part of a nonzero triple but *positive*, so the relation really has degree 2; this comes from the shape of the product, since $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ 1 & b \end{smallmatrix}\right) = \left(\begin{smallmatrix} q & p+qb \\ s & r+sb \end{smallmatrix}\right)$ — the new bottom-left entry is the old bottom-right one, so a positive bottom row stays positive. For the transport to α itself the leading coefficient is the form (r, s, t) evaluated at the rational point $-S/R$, and that cannot vanish because the roots of (r, s, t) are α_m and its conjugate, neither of which is rational.

Theorem 40 (Lagrange, 1770)**.** *The continued fraction of every quadratic irrational is eventually periodic.*

Proof. Write the complete quotients as $\alpha_k = (P_k + \sqrt{D})/Q_k$ with $Q_k \mid D - P_k^2$; the recurrence $P_{k+1} = a_k Q_k - P_k$, $Q_{k+1} = (D - P_{k+1}^2)/Q_k$ preserves that divisibility, since $P_{k+1} \equiv -P_k \pmod{Q_k}$. Call a state *reduced* when $\alpha_k > 1$ and $-1 < \bar{\alpha}_k < 0$. For a fixed D there are only finitely many reduced states, and on them the recurrence is invertible, so once the orbit is reduced it is *purely* periodic — that is the argument of `palindrome.tex` §2–3, which never used $Q = 1$. It is therefore enough that some α_k be reduced.

For $k \geq 1$ we have $\alpha_k > 1$, so $a_k = \lfloor \alpha_k \rfloor \geq 1$. Conjugating $\alpha_k = a_k + 1/\alpha_{k+1}$ gives $\bar{\alpha}_k = a_k + 1/\bar{\alpha}_{k+1}$. Hence if $\bar{\alpha}_k < 0$ then $1/\bar{\alpha}_{k+1} = \bar{\alpha}_k - a_k < -1$, so $-1 < \bar{\alpha}_{k+1} < 0$ and α_{k+1} is reduced.

So suppose no α_k is reduced. Then $\bar{\alpha}_k > 0$ for every $k \geq 1$ — it is never 0, as $\sqrt{D}$ is irrational. Positivity of $\bar{\alpha}_{k+1} = 1/(\bar{\alpha}_k - a_k)$ forces $\bar{\alpha}_k > a_k \geq 1$, so α_k *and* $\bar{\alpha}_k$ exceed 1 for every $k \geq 1$. Subtracting the two conjugate recurrences, and using $\alpha_k - a_k = 1/\alpha_{k+1}$ and $\bar{\alpha}_k - a_k = 1/\bar{\alpha}_{k+1}$,

$$\alpha_{k+1} - \bar{\alpha}_{k+1} \;=\; \frac{1}{\alpha_k - a_k} - \frac{1}{\bar{\alpha}_k - a_k} \;=\; (\bar{\alpha}_k - \alpha_k)\,\alpha_{k+1}\bar{\alpha}_{k+1}.$$

Since $\alpha_{k+1}\bar{\alpha}_{k+1} > 1$, the gap $|\alpha_k - \bar{\alpha}_k|$ is strictly increasing. But $\alpha_k - \bar{\alpha}_k = 2\sqrt{D}/Q_k$, so $|Q_{k+1}| < |Q_k|$ for every $k \geq 1$: a strictly decreasing sequence of positive integers. That is impossible. □

Two remarks. First, $\alpha_k\bar{\alpha}_k = -Q_{k-1}/Q_k$, so for $k \geq 1$ (where $\alpha_k > 0$) the condition $\bar{\alpha}_k < 0$ says exactly that Q_{k-1} and Q_k have the *same sign*. That is Lagrange's own criterion, *Additions* art. 41: "lorsque dans la série des nombres $E, E_1, E_2, \ldots$ on en trouvera deux consécutifs de même signe, celui des deux qui sera moindre que $\sqrt{B}$ sera déjà nécessairement périodique."

Second, the textbook proof is different: it passes through the convergents p_k/q_k, the bound $|\alpha - p_k/q_k| < 1/q_k^2$, and either the invariance of the discriminant of $A_k x^2 + B_k x + C_k$ or the uniqueness of the continued-fraction expansion. The descent above needs none of that, which is why the formalisation is short.

This is `lean/Quadratic.lean`: `exists_reduced` is the preperiod, `eventually_periodic` the theorem. One thing the integer recurrence does need there and did not for $\sqrt{n}$: $\lfloor (P + \sqrt{D})/Q \rfloor$ equals $\lfloor (P + a_0)/Q \rfloor$ only when $Q > 0$, and for a general quadratic irrational the Q_k change sign — as the first remark says, they must. For $Q < 0$ the value is $-\lfloor (P + a_0)/(-Q) \rfloor - 1$.

11 What is not proved here

Nothing. Both halves of Lagrange's theorem are proved and formalised. Euler's is §10, in `lean/Lagrange.lean`; Lagrange's own is Theorem 40, in `lean/Quadratic.lean`.

The numbers in the tables of `sturmian.tex` — the slopes, the four derived sequences of $n = 7$, the gap lists for $n = 2$ and $n = 7$, the four factors of length 3 — have been checked in exact arithmetic in $\mathbb{Q}(\sqrt{n})$.